

Heuristics

Michael D. Mumford
University of Oklahoma, Norman, Oklahoma, USA

Lyle E. Leritz
University of Oklahoma, Norman, Oklahoma, USA

Glossary

correlation A measure of the strength of the relationship between two measures.
divergent thinking tests Open-ended problem-solving tasks in which people are asked to generate multiple solutions.
domain An area of work or task performance involving a shared body of knowledge and skills.
heuristics Strategies for executing the processing of operations involved in complex problem solving.
ill-defined problems Situations in which the goals are unclear and multiple paths to problem solutions are available.
problem construction The cognitive process involved in structuring, or defining, the problem to be solved.
processes Major cognitive operations that must be executed during problem solving.
scenario A written description of a problem and its setting.
think aloud A procedure for analyzing problem solving based on verbalizations during task performance.

The strategies people use to solve complex problems are commonly referred to as heuristics. The particular heuristics people apply in problem-solving, and the effectiveness with which they execute these heuristics, represent critical influences on performance in a variety of settings. In the present article, the procedures used to identify the heuristics involved in performance and the procedures used to measure the effectiveness of heuristic application are examined.

Introduction

History

The term "heuristics" was first applied in the social sciences some 50 years ago. Initially, this term was used to refer to the strategies people employed to reduce the cognitive demand associated with certain decision-making tasks. These strategies involved, for example, "satisficing," which refers to peoples' tendency to use readily available representations as a basis for framing decision tasks. "Means-end analysis" was the term coined to describe a strategy whereby people work backward from a given goal using trial and error to identify the operations needed for problem solving.

As interest grew in problem-solving, planning, and decision-making on the complex, ill-defined cognitive tasks encountered in the real-world (such as planning the Olympic games, creating a new aircraft, or selecting an investment portfolio), it became apparent that multiple solution paths exist that might lead to successful performance. Alternative solution paths, often paths in which simplification proves useful, allow for multiple alternative strategies that might contribute to performance. Accordingly, the concept of heuristics was expanded, and the term is now commonly used to describe both effective and ineffective strategies people apply in executing complex cognitive processing operations, although some scholars prefer to limit use of this term to strategies that simplify complex cognitive operations.

Illustrations and Applications

The shift in conceptualization of cognitive processing has led to a new wave of research intended to identify the various heuristics linked to good and poor performance on different kinds of complex cognitive tasks. In one study along these lines, there was an attempt to identify the heuristics related to performance when people are gathering information for use in creative problem-solving. It was found that better performance was observed on

creative problem-solving tasks when people searched for key facts and anomalies rather than for a wide array of information. In another study along these lines, the researchers sought to identify the heuristics contributing to performance on managerial planning tasks. They found that performance improved when plans were structured around a limited number of key causes—specifically, key causes under ready managerial control. These illustrations of recent research are noteworthy in part because they illustrate one reason why social scientists are interested in heuristics. By identifying the heuristics associated with good and poor performance, it becomes possible to identify the kind of interventions that might be used to improve performance. In fact, studies of heuristics have provided a basis for job redesign efforts, software development, reconfiguration of control systems, and the design of new educational curriculum. Moreover, studies of heuristics have provided a new way for looking at, and assessing, complex cognitive skills.

Although few would dispute the importance of these practical applications, studies of heuristics have proved even more important for theoretical work. Studies of heuristics have not only allowed validation of models of complex processing operations, they have allowed social scientists to specify how various processes are executed in certain performance domains. Indeed, many advances in theories of problem-solving, decision-making, and planning can be traced to identification of heuristics associated with more or less effective execution of certain key cognitive processes.

Objective

Despite the theoretical importance of heuristics, the practical implications of studies of heuristics beg a question: How is it possible to go about identifying relevant heuristics and measuring their application? The intent here is to examine the relative strengths and weaknesses of the various approaches that have been used to identify, and measure, the heuristics people apply to tasks calling for complex cognitive processing activities. More specifically, three general approaches that have been applied are examined: observational, experimental, and psychometric. In examining the methods applied in each of these three approaches, there is no attempt to provide a comprehensive review of all pertinent studies. Instead, the general approach is described and illustrated through select example studies.

Observational Methods

Naturalistic Observations

Although they are not commonly used to assess differential effectiveness, naturalistic observations are used to identify the heuristics people apply as they work on real-world problems, primarily complex problems in which performance unfolds over a long period of time. Investigator observations represent one of the more frequently used observational techniques. In one example of how this technique has been applied in studies of heuristics, research meetings in six microbiology laboratories varying in productivity were recorded. The video and audio recordings, along with relevant documents and notes, were then analyzed for content indicative of select cognitive operations identified in prior research. This content analysis, focusing on problem-solving behaviors, led to the identification of a number of heuristics that appeared to contribute to scientific problem-solving, such as a focus on explanation of inconsistent results and the application of local, or near, analogies in formulating problem solutions.

Observations require ongoing access to people working on a problem. When ongoing access cannot be obtained, naturalistic investigators commonly turn to an alternative technique for identifying heuristics, the interview. In interviews intended to identify heuristics, questions are developed that focus on the actions taken in producing a piece of work. Interviewees' recall of their activities is used to draw inferences about the heuristics being applied. For example, in one study, the interest was in the heuristics used by artists as they developed, over a period of months, pieces of work for exhibition or commission. Sixteen artists were interviewed for 1 or 2 hours on three occasions: (1) as they began work on the project, (2) midway through the project, and (3) after the project was completed. A structured interview format was applied such that these artists were asked to describe the specific actions they took during each phase of work on the project. A content analysis of the resulting interview data indicated that, during idea development, artists were not concerned simply with expression. Instead, they structured ideas, expanded initial ideas, and restructured these ideas before they began work on the piece.

In contrast to direct observations, studies seeking to identify the heuristics applied to complex real-world tasks have, from time to time, used an indirect observational approach based on analysis of historic records. Use of record-based approaches has been most common in studies of scientists, engineers, musicians, and writers, who work in fields in which a substantial body of written material (e.g., articles, drafts, notes) is typically produced. Content analysis of this material, especially analysis of changes in peoples' strategies over time as they move toward completion of a project, is used to draw inferences about the heuristics involved in the work. One illustration of this approach involved an historic analysis of the heuristics used by Bell and Edison in their attempts to develop the first working telephone. Although the need for a sizable body of records tracking progress on a project has limited application of this technique to select

fields, the widespread use of computers may, in the future, permit application of historic techniques in a wider range of domains.

Structured Observations

All unstructured observational techniques suffer, to some extent, from the problem of selective information presentation in the presence, or assumed presence, of observers. However, a more critical problem lies in the fact that these techniques do not capture data about heuristics that remain unarticulated. To address this problem, many researchers have come to rely on "think-aloud" protocols. In this technique, people are asked to verbalize, or think aloud, as they work on a cognitive task. Transcripts, or recordings, of these verbalizations are obtained. Heuristics are identified through a content analysis focusing on verbalizations about strategy selection, justifications for the actions being taken, and the sequences of actions described. This approach has been used to identify the heuristics used by anesthesiologists planning medical operations and to identify the heuristics used by students working on creative problem-solving tasks.

The widespread use of think-aloud procedures in heuristic identification may be traced to three advantageous characteristics of this technique. First, protocols can be obtained for people who presumably are more or less skilled with respect to the application of relevant heuristics—experts versus novices, successful versus unsuccessful problem solvers, and gifted versus nongifted students. Comparison of these groups, of course, allows relatively unambiguous identification of the heuristics associated with good and poor performance. Moreover, vis-à-vis careful structuring of the conditions of task performance, situational factors moderating heuristic application can be identified. Second, because all groups are exposed to a common task, or set of tasks, controlled inferences about heuristic application are possible. Third, content coding schemes can be tailored to specific actions on a known set of problems, thereby enhancing the reliability and validity of assessments.

Although these observations recommend the application of think-aloud procedures in identifying heuristics, this technique does have limitations. Due to the nature of verbalization techniques, it is difficult to apply this approach to highly complex real-world tasks, especially tasks for which performance unfolds over substantial periods of time. Moreover, unless multiple, carefully selected, stimulus tasks are applied, the generality of the conclusions flowing from application of this technique are open to question. Along related lines, it is difficult to collect protocol information for a large number of people, a characteristic of the technique that limits power and population inferences. Finally, due to the complexity of the material obtained, it is difficult to apply this technique in appraising differences among people in the effectiveness of heuristic application.

Scenario-based measures represent an alternative technique used to examine the effectiveness of heuristic application. Scenario assessments present people with written descriptions of a complex real-world cognitive task. Subsequently, people are asked to respond, in writing, to a series of questions about the kinds of actions they would take to address the problem broached in a scenario. The questions asked following a scenario are intended to elicit the exhibition of certain heuristics. Judges' evaluations of either the nature, or effectiveness, of heuristic application provide a basis for assessment.

One illustration of the scenario technique may be found in a study in which 1818 army officers were presented with a set of military planning scenarios calling for creative problem-solving. Subsequently, these officers were asked to provide short answers, in writing, to a series of questions intended to elicit heuristic application. Questions asked, for example, what information should be gathered in planning an invasion, or what would be key problems once the invasion began. Judges' evaluations of responses to these questions with respect to the effective application of the relevant heuristics produced interrater agreement coefficients near 0.70. More centrally, these measures of heuristic application were found to yield multiple correlations in the 0.40s when used to predict various measures of on-the-job performance.

Experimental Methods

Experimental techniques are used in both the identification of heuristics and the assessments of heuristic application. Experimental techniques base identification and assessment of heuristics on observable performances on a particular task or set of tasks. Four general kinds of tasks are commonly applied in studies of heuristics based on the experimental approach: optimization tasks, choice tasks, skill acquisition tasks, and simulation/gaming tasks.

Optimization Tasks

The basic principle underlying optimization tasks is that departures from an optimal standard can be used to appraise the heuristics associated with sub par performance. Some studies apply a variation on this approach whereby departures from a theoretical minimum are used to identify the heuristics that contribute to performance. Accordingly, application of this approach, an approach commonly applied in studies of decision-making heuristics, is contingent on the feasibility of determining theoretical minimums and maximums.

One illustration of this approach may be found in a study that used computational modeling to define best and worst possible performance on a medical diagnosis task. The maximum possible performance was specified using statistical decision theory. The worst possible performance was defined based on a randomization model. Six heuristics involved in medial diagnosis were examined. It was found that performance approached the statistical optimum when a schema, or principle-based search, was applied in information acquisition, and multiple hypotheses were generated concerning the cause of the illness.

Of course, when optimization techniques are applied, the kind of heuristics identified will depend on the standards, either minimum or maximum, being used as benchmarks. Heuristics associated with suboptimal performance, biases, or errors will be identified when maximum standards are applied. Heuristics contributing to enhanced performance will be identified when minimum standards are applied. Though the referencing of heuristics against *a priori* theoretical standards is an attractive feature of this approach, the conclusions drawn in optimization studies will depend on the validity of the minimums and/or maximums applied. As a result, when strong *a priori* theory cannot be used to define applicable standards, it will prove difficult to apply this approach.

Choice Tasks

Choice tasks, in contrast to optimization tasks, apply relative, rather than absolute, standards to measure heuristics. More specifically, choice tasks measure heuristics by examining preferences for applying one heuristic over another as people work on certain cognitive tasks. Assessment is based on the assumption that people will use preferred heuristics in performing both the task at hand and in performing other tasks lying in the same domain. As a result, use of this approach typically requires prior identification of relevant heuristics. It is applied when the concern at hand is identifying more or less useful heuristics. When these conditions are met, however, the ability to link heuristic preferences to performance on a variety of tasks makes this approach an attractive vehicle for assessing heuristic application.

This approach has been used to assess the heuristics involved in problem construction, a processing operation commonly held to play a key role in creative problem-solving. In one study, people were presented with a short written description of four scenarios in which the problem at hand could be defined in a number of different ways. People were to review 16 alternative problem definitions and were asked to select the best four definitions under conditions in which these alternatives were structured to reflect the tendency to define problems in terms of (a) goals, (b) procedures, (c) restrictions, and (d) key information. It was found that most people tended to define problems in terms of key information. However, performance on two complex creative problem-solving tasks was related to a preference for defining problems in terms of procedures and restrictions. Thus, people apparently preferred to use a heuristic that was not one of those contributing to subsequent performance.

Skill Acquisition Tasks

When prior research has provided strong theory concerning the heuristics underlying performance, an alternative to choice tasks may be found in skill acquisition tasks. The assumption underlying the skill acquisition approach is that differences among people in heuristic application can be induced through training. Thus, this approach is applied when heuristics can be developed, typically through relatively short-term interventions. Subsequent comparison of trained and untrained groups with respect to performance on one or more transfer tasks provides the basis for identifying the heuristics contributing to performance.

This approach was used to examine three heuristics held to be involved in creative problem-solving: (1) a hierarchical heuristic whereby ideas are generated by looking for communalities among concepts, (2) a brainstorming heuristic whereby performance is held to depend on generation of many ideas, and (3) a changing perspectives heuristic whereby ideas are generated by considering the views of others. Undergraduates were trained in the application of one of these heuristics, and their ability to generate novel solutions to a roommate interaction problem was assessed. It was found that the hierarchical heuristic resulted in significantly better solutions on this creative problem-solving task as compared to the brainstorming and changing-perspectives heuristics.

Not only can skill acquisition tasks be used to identify the heuristics contributing to performance, variations of this approach can be used to assess the effectiveness of heuristic application. For example, in one study, people were trained in the application of various heuristics involved in the processing activities required for creative problem-solving. The rate of heuristic application and the knowledge about relevant heuristics were assessed. It was found that these measures evidenced adequate validity in performance prediction, suggesting that learning may at times provide a basis for assessing heuristic application.

Simulations/Gaming Tasks

The fourth, and final, experimental procedure used in the identification and assessment of heuristics is simulation or gaming. In simulation and gaming exercises, heuristics are

identified and measured through the choices people make as they work on the exercise. In one study, a record was kept of operating decisions and requests for information as managers worked on a simulation exercise intended to model the operations of a small manufacturing company. It was found that the level of profits, a criterion measured through quarterly profits, was associated with a smaller number of decisions and a tendency to focus decision-making on a limited number of key issues linked to efficiency considerations.

Though simulations and gaming exercises provide an appealing tool for identifying and measuring heuristics, three points should be borne in mind when applying this approach. First, simulation and gaming exercises will prove useful only if decision and action metrics are formulated in such a way that they capture relevant heuristics. Second, the validity of the resulting assessments will depend on these metrics as much as the apparent realism of the exercise. Third, when these exercises are applied, some attempt should be made to control for gaming heuristics, i.e., heuristics specific to the exercise that are not relevant to the general form of task performance.

Psychometric Methods

In the psychometric approach, heuristics are identified and assessed with respect to people, and the performance differences observed among people, rather than with respect to the particular actions observed for a given set of tasks. This point is of some importance because the psychometric approach tends to emphasize heuristics associated with performance differences, discounting heuristics that are applied in a similar fashion by all individuals. To elicit these individual differences, psychometric studies rely on one of two basic techniques: self-report and tests.

Self-Report

One version of the self-report approach assumes that people are aware of, and understand the implications of, the strategies they apply as they execute tasks lying in different domains. Of course, given this assumption, it is possible to identify heuristics simply by asking people about the heuristics they apply in different endeavors. In accordance with this proposition, in one study, undergraduates were asked to indicate whether they applied heuristics such as visualization, step-by-step analysis, and analogies when working on interpersonal, academic, or daily life tasks. It was found that people could describe where, and how frequently, they applied these general heuristics. This direct reporting approach, however, has proved less effective when specific heuristics, particularly heuristics applied nearly automatically, are under consideration.

When it is open to question whether people can say when, and how, they apply heuristics, an alternative strategy is often used. In an approach referred to as the critical-incidents technique, people are not asked to report on strategies applied. Instead, they are asked to indicate the behaviors they exhibited when performing certain tasks. Specifically, they are asked to describe the setting, the event prompting performance, the actions taken in response to the event, and the outcomes of these actions. Content analysis comparing good and poor performers, or good and poor performances, is used to identify more or less effective performance strategies and the behaviors associated with application of these strategies. Assessment occurs by having people indicate either the frequency or extent to which they manifest behaviors associated with an identified heuristic.

Testing

In testing, people are not asked to describe heuristics or the behaviors associated with these heuristics. Instead, the capability for applying heuristics is inferred based on peoples' responses to a series of test items. In one variation on this approach, the objective scoring approach, test items are developed such that the problems presented call for certain processing activities. Response options are structured to capture the application of heuristics linked to effective process application. In one study, test items were developed to measure selective encoding, selective comparison, and selective combination. It was found that gifted students differed from nongifted students in that they were better able to identify relevant information (selective encoding).

In the subjective scoring approach, problems (typically open-ended problems) are developed in a way that a variety of responses might be used to address the problem. Judges are then asked to review peoples' responses and assess the extent to which they reflect the application of certain heuristics. This approach is applied in scoring the divergent thinking tests commonly used to measure creativity by having judges evaluate responses with respect to three creative strategies: generating a number of ideas (fluency), generating unusual ideas (originality), and generating ideas through the use of multiple concepts (flexibility). In another illustration of this approach, army officers were asked to list the changes in peoples' lives that might happen if certain events occurred (e.g., What would happen if the sea level rose?). Judges scored these responses for heuristics such as use of a longer time frame, application of principles, and a focus on positive versus negative consequences. Use of longer time frames and application of principles were found to be related to both performance on managerial problem-solving tasks and indices of real-world leader performance, yielding multiple correlations in the 0.40s.

Conclusions

Clearly, a variety of procedures have been developed for identification and assessment of the heuristics people use in working on complex, ill-defined problems. Moreover, these procedures have proved useful in identifying heuristics, and measuring heuristic application, across a number of performance domains calling for complex cognition. What should be recognized, however, is that all of these approaches evidence certain strengths and weaknesses. The observational approach has typically proved most useful in heuristic identification. Experimental and psychometric methods are more commonly used to identify the impact of heuristics on performance and measure individual differences in heuristic application. Along similar lines, certain techniques subsumed under these three basic methods appear, by virtue of their assumptions, more appropriate than are others for addressing some questions, and measuring some heuristics. These observations are of some importance because they suggest that a comprehensive understanding of the heuristics involved in a certain type of performance will require a multimethod, multitechnique approach.

Although the kind of multimethod, multitechnique studies called for are not a simple undertaking, the evidence indicates that work along these lines will be worth the time and effort. Not only has identification of the heuristics involved in high-level cognitive performance proved critical in theory development, the information provided by these studies has provided new tools for developing and assessing peoples' performance capacities. Given the foundation of human performance in complex cognitive activities such as planning, decision making, and creative problem-solving, it can be expected that heuristic measurement will become an increasingly important aspect of social measurement over the course of the 21st century.

See Also the Following Articles

Intelligence Testing • Problem-Solving Methodologies • Psychological Testing, Overview

Further Reading

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